User Preference Adaptation in Human-Robot Interaction Using Reinforcement Learning: A Review of Personalization Approaches

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***Abstract*—Human-Robot Interaction (HRI) is increasingly moving toward personalized and user-centered systems capable of adapting their behavior to individual user needs, preferences, emotional states, and comfort levels. Traditional robotic systems are typically designed to optimize task completion, efficiency, and safety; however, they often fail to account for the diversity of human preferences and behavioral responses during interaction. Reinforcement Learning (RL) has emerged as a promising approach for enabling robots to learn adaptive behaviors through interaction and feedback, allowing them to personalize their actions while maintaining task effectiveness. This report reviews several recent approaches that utilize reinforcement learning and adaptive user modeling to support personalization in HRI. The reviewed studies include Preference-based Action Representation Learning (PbARL), dynamic user adaptation systems, comfort-aware reinforcement learning frameworks, and adaptive socially assistive robots for cognitive rehabilitation. These approaches demonstrate how robots can adapt to user preferences, engagement levels, communication styles, emotional states, and comfort requirements through learned policies and personalized interaction strategies. The findings indicate that reinforcement learning can significantly improve personalization, user satisfaction, engagement, and interaction quality while preserving task performance. The report also discusses the strengths, limitations, and future research directions of current personalization techniques, highlighting the importance of developing intelligent robots capable of long-term adaptation and user-centered interaction.**

***Index Terms*—Human-Robot Interaction, Reinforcement Learning, Personalization, User Preference Adaptation, Adaptive Behavior, Socially Assistive Robots, Preference Learning, Intelligent Robotic Systems**

I. Introduction

Human-Robot Interaction (HRI) has become an important research area as robots are increasingly deployed in environments where they interact directly with humans, including healthcare, education, assistive living, rehabilitation, and industrial settings. In such environments, successful interaction depends not only on the robot's ability to perform tasks accurately but also on its ability to adapt its behavior to the individual characteristics, preferences, and needs of different users. Since people vary in their comfort levels, communication styles, emotional responses, cognitive abilities, and behavioral expectations, a one-size-fits-all interaction strategy is often insufficient for creating effective and satisfying human-robot relationships.

Personalization has therefore emerged as a key requirement for next-generation robotic systems. Personalized robots can modify their behavior according to user preferences, maintain engagement during long-term interactions, and provide more natural and socially acceptable experiences. However, designing adaptive behaviors manually is challenging because user preferences are often complex, dynamic, and difficult to model explicitly. Furthermore, excessive adaptation may negatively affect task performance if robots prioritize user preferences at the expense of accomplishing their intended objectives.

Reinforcement Learning (RL) provides a promising solution to this challenge by enabling robots to learn optimal behaviors through interaction with their environment and users. Rather than relying on predefined rules, RL-based systems can continuously improve their decision-making policies by observing user responses and receiving feedback. Recent advances in preference-based reinforcement learning, adaptive user modeling, comfort-aware learning, and socially assistive robotics have demonstrated the potential of RL to personalize robot behavior while maintaining efficiency and task effectiveness. For example, the Preference-based Action Representation Learning (PbARL) framework proposed in [1] enables robots to adapt their actions according to individual user preferences while preserving the task knowledge acquired through pretrained policies. Similarly, the adaptive interaction framework presented in [2] dynamically adjusts robot behavior based on user attention, communication patterns, and learning performance to improve engagement during interactions. Human-centered adaptation has also been explored through comfort-aware reinforcement learning, where robot trajectories and movement behaviors are adjusted according to user comfort preferences [3]. In the domain of socially assistive robotics, reinforcement learning has been employed to personalize verbal and non-verbal robot behaviors for older adults with Mild Cognitive Impairment (MCI), aiming to sustain engagement during cognitive training activities [4].

This report reviews several representative studies that address user preference adaptation in HRI using reinforcement learning and adaptive interaction strategies. The selected papers explore different dimensions of personalization, including preference-aware action representation learning [1], dynamic adaptation based on user behavior and engagement [2], comfort-aware robot motion planning [3], and adaptive socially assistive robots for cognitive training [4]. By analyzing these approaches, the report aims to provide an overview of current techniques, identify common challenges, and highlight future directions for developing intelligent robotic systems capable of long-term personalized interaction.

II. Overview of Selected Papers

A. Personalization in HumanRobot Interaction through Preferencebased Action Representation Learning (PbARL)

The authors in paper [1] addresses the challenge of personalization in HumanRobot Interaction (HRI), where different users often prefer different robot behaviors even when the robot is performing the same task. Existing Preferencebased Reinforcement Learning (PbRL) approaches personalize robot behavior by collecting human preference feedback and training a personalized policy from scratch. However, this process is inefficient because human feedback must simultaneously teach both task execution skills and individual user preferences, resulting in a large feedback requirement that limits practical deployment. Furthermore, directly finetuning pretrained robot policies using preference information can degrade task performance, causing failures in task completion and reducing user satisfaction. Consequently, the paper investigates how robots can efficiently adapt to individual user preferences while preserving the strong task performance already learned by pretrained policies.

To address this problem, the authors propose a novel framework called Preferencebased Action Representation Learning (PbARL). Rather than retraining or directly finetuning a robot policy, PbARL keeps the pretrained policy frozen and learns a new latent action representation that aligns robot actions with user preferences. The approach begins with a pretrained policy that has already learned effective task execution and interaction skills, serving as the source domain. Human feedback is then collected through preference comparisons between different robot behaviors, which are used to train a preferencealigned reward model capable of estimating the actions preferred by a user. The key innovation of the framework is a Mutual Information Action Encoder implemented using a Conditional Variational Autoencoder (cVAE). This encoder learns a latent action space that simultaneously preserves original task performance and adapts behavior according to user preferences. To achieve this balance, the model maximizes mutual information between the source domain, represented by the pretrained policy behavior, and the target domain, represented by userpreferred behavior. PbARL further employs a multiloss optimization strategy consisting of four objectives: Reconstruction Loss, which preserves the original action structure; Preference Loss, which aligns actions with user preferences; KL Divergence Loss, which regularizes the latent space; and Dynamic Loss, which ensures consistency with state transitions and improves action controllability. The framework was evaluated on the Assistive Gym benchmark using three assistive tasksfeeding, drinking, and itch scratchingand was additionally validated through a realworld user study involving a Kinova Jaco robotic arm and eight participants.

The experimental results demonstrate that PbARL achieves superior personalization while maintaining strong task performance. Across all evaluated tasks and user types, PbARL consistently obtained higher preference rewards than the original pretrained policy while maintaining substantially higher task success rates than conventional PbRL approaches. Preference FineTuning (PrefFT) improved personalization but led to significant reductions in task success, whereas traditional PbRL methods required considerably more human feedback and performed poorly when feedback was limited. For example, in the feeding task for neutral users, the pretrained policy achieved a 92% success rate with a preference reward of 11.2, PbRLHRI achieved a 41% success rate with a preference reward of 12.1, and PrefFT achieved a 73% success rate with a preference reward of 45.6. In comparison, PbARL achieved an 88% success rate and the highest preference reward of 50.7, demonstrating the best balance between personalization and successful task completion. The realworld user study further showed that participants rated PbARL significantly higher in terms of personalization and overall satisfaction compared with both the pretrained policy and Preference FineTuning. Statistical analysis confirmed that these improvements were significant ($p < 0.01$), indicating that users preferred robots adapted through PbARL because they remained effective while exhibiting behaviors aligned with individual preferences.

The proposed framework offers several notable strengths. First, it efficiently utilizes human feedback by leveraging existing pretrained policies instead of learning from scratch, thereby reducing the amount of preference feedback required. Second, by keeping the original policy frozen, it preserves task performance and avoids the degradation commonly observed in finetuning approaches. Third, the framework demonstrates strong personalization capabilities, successfully adapting robot behavior to different user preferences while maintaining task effectiveness. Fourth, PbARL is broadly applicable and can be integrated with robot policies trained using standard reinforcement learning methods. Fifth, the inclusion of both simulated benchmark experiments and realworld user evaluations strengthens the validity and practical relevance of the findings. Finally, the paper introduces a novel perspective by formulating personalization as a mutualinformation maximization problem between task knowledge and user preferences.

Despite these advantages, the study has several limitations. The realworld evaluation involved only eight participants, limiting the generalizability of the results. Additionally, the experiments focused exclusively on three assistive tasksfeeding, drinking, and itch scratchingleaving the framework's effectiveness on more complex or diverse tasks unexplored. The approach also depends on the availability of highqual-

ity pretrained policies, meaning that poorly trained policies may negatively affect personalization outcomes. Furthermore, performance is sensitive to hyperparameters such as action list size, loss weights, and latentspace configuration, and the authors acknowledge that determining the optimal action list size requires further investigation. Finally, the framework models preferences using collected feedback but does not support continuously evolving user preferences over longterm interactions, making lifelong personalization an important direction for future research.

Overall, the paper presents PbARL as a novel and effective personalization framework for HumanRobot Interaction that learns userpreferenceaware action representations while preserving the capabilities of pretrained robot policies. By maximizing mutual information between task knowledge and user preferences, the framework achieves superior personalization, higher user satisfaction, and stronger task success rates than existing Preferencebased Reinforcement Learning and preference finetuning approaches. The work represents a significant contribution toward practical, efficient, and user-centered robotic assistants capable of adapting to individual user needs without sacrificing performance.

B. Towards adaptation and personalization in task based on Human-Robot Interaction

The study focuses on one of the key challenges in Human-Robot Interaction (HRI) while maintaining user engagement during long-term interactions [2]. While social robots often attract users initially because of their novelty, user interest and attention frequently decline over time as the novelty effect diminishes. The authors argue that many existing adaptive systems rely on static user models and provide limited personalization capabilities, making it difficult for robots to sustain meaningful interactions over extended periods. In addition, many systems fail to account for important human factors such as emotions, communication styles, attention levels, and individual preferences. To address these limitations, the paper investigates whether a robot can dynamically adapt its behavior and personalize conversations based on user characteristics and real-time interaction cues in order to increase engagement and improve interaction quality. The authors hypothesize that combining personalized conversations with adaptive robot behavior can help maintain user interest and foster stronger long-term human-robot relationships.

To achieve this objective, the researchers developed a Dynamic User Adaptation System integrated into a humanoid NAO robot. The proposed framework combines user modeling, machine learning techniques, behavioral adaptation, and personalized dialogue generation. The system architecture consists of four major components: operational parameters, a decision-making module, a user model, and a user interface supported by user indicators. The robot continuously monitors three categories of user behavior. The first category is attention, which is measured through face gaze direction and user posture to determine whether the user remains focused on the activity. The second category is communication quality, which is assessed through the number of spoken words and detected facial emotions to evaluate how actively the user participates in the interaction. The third category is learning performance, which is measured using correct and incorrect answers as well as response time to determine how well users understand and engage with the task. Several artificial intelligence techniques are incorporated into the framework, including a Haar Cascade Classifier for face detection and gaze monitoring, a MobileNet deep neural network for facial emotion recognition, Levenshtein Edit Distance for evaluating verbal responses, and a rule-based decision-making system for behavioral adaptation. In addition, the user model stores personal information such as the user's name, birthday, school year, favorite sport, favorite food, favorite music, favorite game, hobbies, and interests. This information is later integrated into conversations to create familiarity and strengthen rapport between the robot and the user.

The framework was evaluated through a storytelling activity designed to assess the effectiveness of adaptation and personalization. During the activity, the robot first collected personal information from participants, then narrated stories and asked questions while dynamically adapting its behavior based on user responses. Personalized conversations related to participants' preferences were integrated throughout the interaction. The experiment involved ten university students with a mean age of 24.42 years, none of whom had prior experience interacting with the NAO robot. Each interaction session lasted approximately twelve minutes. The results demonstrated that the robot successfully modified its behavioral profile during interactions by continuously monitoring attention, communication, and task-performance indicators. Based on these observations, the system adjusted communication intensity and gestural behavior to better suit the user's engagement level.

To assess user perceptions, participants completed questionnaires after the interaction sessions. The findings indicated that users generally enjoyed the activity and were able to perceive the robot's personalization efforts. Participants also recognized the value of adaptive and personalized robotic systems and believed that personalization contributes to stronger human-robot relationships. The emotion recognition module produced acceptable results when users displayed clear emotional expressions. However, the researchers observed that concentrated or focused facial expressions were frequently misclassified as negative emotions such as sadness, disgust, or fear. Despite this limitation, the overall results supported the authors' hypothesis that users could perceive the robot's adaptive behavior and personalization features, suggesting that adaptation can improve user engagement during interactions.

The study presents several notable strengths. First, unlike many existing systems that rely on static user models, the proposed framework continuously adapts robot behavior using real-time user signals. Second, the system enables personalized interactions by storing user information and incorporating it into future conversations, creating more engaging and human-like experiences. Third, the framework employs multimodal user analysis by simultaneously considering attention,

emotions, speech, and task performance, providing a more comprehensive assessment of user engagement. Fourth, the approach was implemented and tested on a real humanoid robot rather than solely in simulation, increasing its practical relevance. Fifth, participants generally appreciated the adaptive and personalized features and recognized their potential value in future human-robot interactions. Finally, the framework supports rapport building and long-term relationship development by remembering user preferences and integrating them into subsequent interactions.

Despite these contributions, the study has several limitations. The evaluation involved only ten participants, limiting the statistical significance and generalizability of the findings. Furthermore, the emotion recognition module frequently misclassified concentration-related facial expressions as negative emotions, reducing the accuracy of behavioral adaptation. The language understanding component also exhibited limitations because the use of Levenshtein Edit Distance sometimes failed to correctly evaluate answers expressed with different wording despite being semantically correct. In addition, the adaptation mechanism primarily relies on rule-based decision-making rather than more advanced learning-based approaches, which may limit flexibility and scalability. Although the framework aims to support long-term interaction, the experimental validation consisted of only a single interaction session, making it difficult to assess long-term effects. Finally, several participants reported difficulties understanding the robot's speech due to limitations in NAO's built-in speech synthesis system.

C. Adaptive Robot Behavior Based on Human Comfort Using Reinforcement Learning

The authors in [3] addressed a growing challenge in HRI as collaborative robots (cobots) become increasingly common in industrial and shared work environments. While many existing robotic systems are optimized for productivity, efficiency, and safety, they often neglect an equally important factor: human comfort. Human comfort during interactions with robots is influenced by several factors, including the distance between the robot and the user, robot movement speed, motion trajectories, perceived safety, and emotional responses. Traditional reinforcement learning approaches generally focus on maximizing task completion performance without considering whether robot behaviors make nearby humans feel comfortable or uncomfortable. Consequently, robots may operate efficiently while simultaneously creating stress or discomfort for users. To address this issue, the authors investigate whether reinforcement learning can be used to train robots that automatically adapt their behavior according to different human comfort levels while still performing their tasks effectively. To answer this question, the researchers designed a configurable training framework in which robot behavior can be directly influenced by human comfort parameters.

To achieve this objective, the authors developed a Comfort Robotic Learning Framework (Comfort-RL) that enables robots to learn task execution while simultaneously adapting their behavior according to user-defined comfort preferences. The framework was implemented using the Unity Simulation Engine, Reinforcement Learning techniques, Stable Baselines3, and ML-Agents. The environment was designed to be publicly available and reproducible, allowing other researchers to replicate and extend the work. The framework supports three robotic tasks: moving to a fixed or random point, moving between multiple points, and picking up a colored ball and placing it on the correct table. Among these, the object manipulation task involving the colored ball was selected as the primary experimental scenario because of its complexity and practical relevance.

The key innovation of the framework is the integration of human comfort directly into the reinforcement learning reward function. Two primary comfort variables were considered. The first is distance-based comfort, which evaluates the distance between the robot and the human user. Lower comfort levels encourage the robot to maintain a greater distance from the user, whereas higher comfort levels permit the robot to move closer. This behavior is modeled using a hyperbolic tangent-based reward function. The second variable is motion-based comfort, which evaluates the smoothness of robot movement by measuring joint angle changes, motion speed, and incremental movement steps. Through this mechanism, robots can learn to move more slowly and predictably when lower comfort thresholds are specified. For policy learning, the researchers employed the Soft Actor-Critic (SAC) algorithm due to its strong performance in continuous action spaces and robotic manipulation tasks. The overall reward function combines task completion rewards with comfort-related penalties and rewards, including both distance and motion considerations. This design encourages the robot to balance efficiency and productivity with human-centered behavioral adaptation.

The framework was evaluated through two primary experiments. In the first experiment, robots were trained without incorporating comfort parameters. The objective was to validate task completion capabilities while testing obstacle avoidance and adaptation to human movement. In the second experiment, robots were trained using ten different comfort levels to measure behavioral changes, analyze trajectory adjustments, and evaluate average human-robot distance alongside task efficiency. Results from the first experiment demonstrated that the robot successfully learned to pick and place objects, avoid collisions, and adapt its trajectories in response to human movement. After approximately 30 million training steps, the framework achieved a task success rate of 90%, demonstrating effective learning and adaptability even when humans moved unpredictably during task execution.

The second experiment revealed significant differences in behavior among comfort-aware models. The results showed that different comfort levels produced noticeably different robot trajectories. Robots maintained larger distances from humans when lower comfort levels were specified, while higher comfort levels resulted in shorter and more direct movement paths. Furthermore, task completion efficiency improved as comfort levels increased. Quantitative analysis

demonstrated that the average robot-human distance decreased by approximately 25% between the lowest and highest comfort settings. Similarly, task completion time, measured by the number of execution steps, decreased by approximately 12% between the lowest and highest comfort levels. Visualizations presented in the paper further illustrated how robot trajectories adapted under different comfort settings. One of the most significant findings of the study was that modifying only a single comfort-related parameter during training was sufficient to generate substantially different robot behaviors without requiring changes to the underlying policy architecture.

The proposed framework offers several important strengths. First, unlike many robotics studies that focus exclusively on task performance, this work explicitly incorporates human comfort into the reinforcement learning process, promoting a more human-centered approach to robot behavior. Second, the framework is highly configurable, allowing researchers to modify tasks, adjust comfort parameters, customize reward functions, and integrate additional robotic systems, thereby increasing its flexibility and research value. Third, the authors released the environment through GitHub, supporting reproducibility and facilitating future community contributions. Fourth, the study combines quantitative performance metrics, distance measurements, execution-time analysis, and visual trajectory evaluations, providing compelling evidence of behavioral adaptation. Fifth, the framework is designed to be scalable across multiple application domains, including manufacturing, healthcare, virtual reality training, and human-assistance robotics. Finally, the robot is trained while interacting with moving humans rather than static environments, increasing the realism and practical relevance of the experiments.

Despite these contributions, the study has several limitations. All experiments were conducted within a simulated environment rather than on physical robotic platforms. Although the framework was designed with future real-world deployment in mind, experiments involving actual robots and human participants have not yet been performed. Additionally, the framework assumes that comfort levels are already available as input values and does not address the challenge of measuring human comfort in real time. The authors suggest that future implementations could estimate comfort through physiological signals such as heart-rate sensors, electroencephalography (EEG) devices, or direct user feedback systems. Furthermore, the comfort model primarily focuses on interpersonal distance and robot motion speed, while other important dimensions of comfort, including trust, anxiety, workload, and emotional state, are not explicitly represented. The evaluation also relied on a single robotic platform and one primary manipulation task, limiting generalization across diverse robotic systems and applications. Another limitation is the high computational cost of training, which required between 12 and 30 million learning steps and multiple days of computation, potentially restricting accessibility for researchers with limited resources. Finally, the effectiveness of comfort-driven behavioral adaptation remains to be validated through real-world experiments involving physical robots and human users.

D. An Adaptive Behaviour-Based Strategy for SARs Interacting with Older Adults with MCI during a Serious Game Scenario

The authors in [4] addresses the challenge of maintaining engagement and motivation among older adults with Mild Cognitive Impairment (MCI) during cognitive rehabilitation activities. MCI is a condition commonly affecting older adults and is characterized by noticeable declines in memory, attention, language, and other cognitive abilities. Although cognitive training programs are often prescribed to slow cognitive deterioration, these interventions frequently involve repetitive exercises that can become monotonous over time. As a result, participants may gradually lose motivation, disengage from training sessions, and eventually discontinue therapy. Socially Assistive Robots (SARs) have emerged as promising tools for supporting cognitive rehabilitation; however, many existing systems rely on static interaction strategies that do not adapt to the user's emotional state, engagement level, or cognitive abilities. This limitation reduces their effectiveness in sustaining attention and fostering meaningful long-term interactions. To address this issue, the authors investigate how a socially assistive robot can dynamically adapt its verbal and non-verbal behaviors to maintain engagement and motivation among older adults with MCI during repetitive cognitive training activities. Specifically, the study explores whether reinforcement learning can be used to identify optimal robot behaviors that encourage users to remain attentive, engaged, and motivated throughout a serious-game-based cognitive training session.

To achieve this objective, the researchers developed an adaptive behavior framework for a Pepper humanoid robot operating within a cognitive serious game environment. The adaptation strategy is based on Q-Learning, a reinforcement learning algorithm that enables the robot to learn which behaviors are most effective at maintaining user engagement. The experimental scenario revolves around a cooking serious game specifically designed for older adults with MCI. The game challenges participants to remember ingredient sequences, recall ingredient weights, answer memory-based questions, and exercise attention and working-memory skills. The interaction is structured into five phases: introduction, recipe presentation, question phase, answer phase, and final feedback phase. Throughout the activity, users communicate primarily with the Pepper robot through voice-based interaction.

To enhance personalization, the researchers implemented two distinct robot personalities. The extroverted personality is characterized by louder speech, a faster speaking rate, higher vocal pitch, larger body gestures, more energetic movements, and highly expressive communication. In contrast, the introverted personality uses a softer voice, smaller gestures, reduced body movement, and a more reserved communication style. The selected personality remains fixed throughout the interaction and is chosen by a caregiver before the session begins. The reinforcement learning model is structured around three key components: state space, action space, and reward

function. The state space consists of several user-related variables, including gaze direction, smiling level, answer correctness, and current engagement level. These indicators are combined to estimate the user's overall state during gameplay. The action space contains three categories of robot responses. Action A0 provides enthusiastic feedback through highly encouraging responses, increased vocal expressiveness, and larger gestures, such as "Awesome! That's the correct answer! You're doing great!" Action A1 provides neutral feedback through standard positive responses and moderate gestures, such as "Good! That's the right answer!" Action A2 delivers stimulating feedback designed to regain user attention and encourage continued engagement, such as "Correct answer! Let's keep up this level of attention!" These actions are further adapted according to the selected robot personality.

The reward function was designed with the primary goal of maximizing user engagement throughout the training process. Higher rewards are assigned when users remain attentive, stay engaged, answer questions correctly, and avoid entering low-engagement states. In this way, the reinforcement learning algorithm seeks to minimize disengagement during repetitive cognitive exercises. To train the system, the authors developed two user models: one representing healthy older adults and another representing older adults with MCI. The robot was trained using 100 epochs and 35 episodes per epoch while employing a Q-learning algorithm with ϵ-greedy exploration. Through repeated interactions, the algorithm learned a policy that maps observed user states to the most appropriate robot actions.

The simulation results demonstrated that the reinforcement learning algorithm successfully converged toward a stable adaptation policy. This indicates that the robot was able to learn which behaviors were most effective for maintaining engagement under different user conditions. The results showed successful learning convergence, meaning that the Q-Learning model gradually identified an optimal behavior strategy and that additional training would provide only marginal improvements. The learned policy enabled the robot to detect changes in user engagement, select actions that increased attentiveness, react differently according to the user's cognitive state, and dynamically adjust both verbal and non-verbal communication strategies. These findings suggest that adaptive behavior can effectively support engagement during repetitive cognitive training tasks, thereby reducing the likelihood of boredom and therapy abandonment.

Conclusion

Personalization is becoming an essential component of modern Human-Robot Interaction systems as robots increasingly operate in environments that require continuous and meaningful interaction with humans. The studies reviewed in this report demonstrate that reinforcement learning and adaptive user modeling provide effective mechanisms for enabling robots to adapt their behavior according to individual user preferences, comfort requirements, engagement levels, and cognitive characteristics. Preference-based Action Representation Learning (PbARL) enables robots to preserve task performance while adapting to user preferences [1], whereas dynamic adaptation frameworks demonstrate the importance of responding to user engagement, communication patterns, and user characteristics during interactions [2]. Similarly, comfort-aware reinforcement learning approaches highlight the value of incorporating human-centered factors such as comfort and proximity directly into the learning process [3], while adaptive socially assistive robotic systems show how reinforcement learning can be used to sustain engagement and motivation among older adults during cognitive training activities [4].

Overall, the reviewed research indicates that adaptive and personalized robot behavior can improve user satisfaction, engagement, interaction quality, and long-term acceptance of robotic systems [1]–[4]. Despite these promising results, several challenges remain, including limited real-world validation, small participant samples, difficulties in accurately modeling human emotions and preferences, and the need to support continuously evolving user preferences over extended interactions. Future research should focus on lifelong personalization, multimodal user modeling, real-time preference estimation, and large-scale real-world deployments. Addressing these challenges will contribute to the development of intelligent, socially aware, and user-centered robotic systems capable of building long-term relationships with diverse users while maintaining robust task performance.

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